

David Braun, Ph.D.

✉ dbraun31@gmail.com

🌐 davebraun.net

in [david-braun-phd](https://www.linkedin.com/in/david-braun-phd)

🔗 [dbraun31](https://github.com/dbraun31)

Professional Summary

Predictive analytics and machine learning specialist with a PhD and 5 years of senior researcher experience designing experiments, analyzing complex datasets, and building predictive models to drive data-informed decisions. Experienced across the full data science lifecycle, from study design and data collection to machine learning, statistical analysis, and results communication. Skilled in collaborating with diverse stakeholders, translating technical findings into actionable insights, and ensuring models are interpretable and valid.

Professional Experience

Senior Researcher

Drexel University ([Neuroscience lab](#)), Philadelphia, PA

Oct 2023 – Present

- Leveraged **unsupervised ML and cluster-based permutation testing** to identify biomarkers of anxiety-related interoceptive dysfunction from 175GB of multimodal EEG/ECG data ([first-authored paper](#); [poster](#); [project blog post](#)).
- Built and maintained **automated Python ETL pipeline** (1,500+ LOC) to convert raw EEG, fMRI, and behavioral data from multiple lab experiments into an industry-standardized format, significantly enhancing reproducibility and cutting manual processing time ([GitHub](#)).
- Applied **novel individual-level PCA and hierarchical clustering** method to uncover unique patterns in spontaneous thinking, advancing diagnostics in mental health and precision medicine ([GitHub](#); [poster](#)).
- Independently developed a Bayesian model investigating neural and cognitive mechanisms underlying adaptive vs. maladaptive forms of off-task thinking for NIH early-career grant proposal ([see Specific Aims](#)).

Senior Researcher

Lehigh University (Cognition lab), Bethlehem, PA

Sep 2020 – Aug 2023

- Led evaluation of COVID-19 non-pharmaceutical interventions (NPIs) using **Bayesian timeseries modeling, random forest and 10,000+ crowdsourced responses**, providing evidence that perceived public compliance with NPIs can improve forecasts 3 weeks ahead ([first-authored paper](#); [project blog post](#); [data/code](#); [poster](#)).
- Designed **deep learning object detection pipeline** to automate video annotation in expert vs. novice eye-tracking studies, improving labeling speed and precision ([paper](#)).
- Validated national survey instrument (N = 1,000+) on public concern for cyber-physical technology using SEM and factor analysis for use in NSF-funded engineering simulation models.
- Supervised and mentored 3 undergraduate researchers on independent projects ([example student poster](#)).

Graduate Researcher

Lehigh University, Bethlehem, PA

Sep 2014 – May 2020

- Developed hierarchical logistic regression models to analyze individual risk preferences in effort-based decision making; first-authored publication cited 35+ times ([first-authored paper](#)).
- Taught R programming to undergraduate and graduate students as lead instructor, improving data literacy and applied statistical training.

Technical Projects

Fraud Detection

Developed and evaluated a fraud prediction model using a public Kaggle dataset, emphasizing smart trade-offs between false positives and business cost. Used feature engineering, cross-validation, and a tuned random forest model with cost-sensitive thresholds to optimize precision-recall tradeoffs ([project blog post](#)).

toBIDS

Created a Python-based ETL pipeline to convert raw neuroscience data (EEG/fMRI) into BIDS standard, enabling

reproducible and automated data workflows ([GitHub](#)).

hPCA

Built an R package implementing hierarchical PCA and clustering to analyze spontaneous thought data at the individual level, aiding in the development of personalized cognitive models ([GitHub](#)).

GradCPTpy

Developed a robust Python implementation of a widely-used cognitive task for attention assessment, integrating experience sampling features. This tool has been adopted by multiple research labs, demonstrating strong usability, reproducibility, and scalability in behavioral data collection and analysis. ([GitHub](#)).

Skills

Programming

- Python (9 yrs; numpy, scikit-learn), R (10 yrs; tidyverse, ggplot2)
- SQL, Git/GitHub, Bash

Machine Learning & Modeling

- Regression, tree-based models, clustering, PCA, deep learning
- Bayesian and frequentist statistics (GLMs, hierarchical models, SEM)
- Causal inference, A/B testing, model interpretability

Data Engineering

- ETL pipeline development, automated workflows
- Data cleaning, BIDS standardization for neuroimaging

Collaboration & Communication

- Cross-functional collaboration with neuroscientists, engineers, and epidemiologists
- 10+ scientific publications, 25+ conference talks
- Mentorship of 5+ undergraduate researchers

Education

Ph.D., Cognitive Neuroscience

Lehigh University

May 2020

M.S., Cognitive Neuroscience

Lehigh University

Dec 2016

B.A., Psychology

Stockton University

May 2013